

Real rank zero does not mean projection density

Disproof of conjecture 00000000920

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Let $A = \mathbb{C}$ with its usual norm and involution $z^* = \bar{z}$. This is an actual homogeneous C^* -algebra:

$$\mathbb{C} \cong C(\{\text{pt}\}, M_1(\mathbb{C})), \quad f \mapsto f(\text{pt}).$$

Evaluation preserves multiplication, involution and norm, since the supremum norm on a singleton equals the norm at that point. In particular, A is AH: it is the inductive limit of the constant homogeneous system $A \xrightarrow{\text{id}} A \xrightarrow{\text{id}} \dots$. No infinite-dimensional restriction appears in the conjecture.

Real rank zero. The self-adjoint elements of A are the real numbers. Given a self-adjoint a and $\varepsilon > 0$, if $a \neq 0$ take $b = a$. It is self-adjoint and invertible, and $\|b - a\| = 0 < \varepsilon$. If $a = 0$, take the real scalar $b = \varepsilon/2$. It is nonzero, hence invertible, and $\|b - a\| = \varepsilon/2 < \varepsilon$. Thus every self-adjoint element is approximable by self-adjoint invertibles, proving real rank zero in precisely the approximation formulation stated in the source.

Projections are not dense. A projection $p \in \mathbb{C}$ satisfies $p^* = p$ and $p^2 = p$. The latter equation gives $p(p - 1) = 0$, so $p = 0$ or $p = 1$. Conversely both are projections. For the self-adjoint element $a = 1/2$,

$$\|p - a\| = \frac{1}{2} \quad \text{for every projection } p.$$

Consequently the open norm ball of radius $1/4$ around $1/2$ contains no projection. The projection semilattice is exactly $\{0, 1\}$, with its usual order, and is therefore not dense among self-adjoint elements.

This disproves the claimed equivalence: the AH algebra has real rank zero but fails the stated density condition. The further cancellation description cannot restore an already-false equivalence.

Interpretation of the density clause. The Chinese version explicitly uses the projection semilattice, while the English version calls it the projection semigroup. Neither notion means arbitrary real linear combinations of projections. Even the more generous interpretation as finite additive sums of projections fails here: such sums are nonnegative integers, each at distance at least $1/2$ from $1/2$. The stable Murray–von Neumann semigroup $V(A)$ is an abstract semigroup of classes, not itself a normed subset of A . The counterexample addresses the literal projection-density assertion in the source. The familiar real-rank-zero characterization by approximation with self-adjoint *finite-spectrum* elements, or by suitable real linear combinations of projections, is a different condition and is not refuted.

Lean certificate. The project constructs the actual continuous-function algebra on a point, its algebra equivalence with $\mathbb{C}$ and the universal property of the constant identity system. It proves self-adjoint invertible approximation, classifies all complex projections, computes the exact norm gap at $1/2$ and proves real rank zero together with failure of projection density.